

MANSFIELD LAHM, OH, USA

WMO: 724200

Lat: 40.820N

Lon: 82.518W

Elev: 1290

StdP: 14.02

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14891

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.7	5.9	-5.9	4.2	2.1	-1.0	5.5	7.2	27.4	37.8	25.6	36.8	8.9	260	0.535

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	88.2	73.1	85.9	71.8	83.6	70.5	75.7	84.5	74.2	82.2	72.8	80.2	9.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	129.0	80.2	71.7	122.9	78.8	70.3	116.9	77.2	40.1	85.0	38.5	82.6	37.2	80.1	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	20.5	18.5	DB	-5.2	91.5	6.3	2.4	-9.7	93.2	-13.4	94.6	-17.0	96.0	-21.6	97.8	(4)
(5)			WB	-5.0	78.0	5.4	1.7	-8.9	79.2	-12.1	80.1	-15.1	81.1	-19.0	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	25.8	28.6	37.4	49.5	60.0	68.8	72.0	70.7	64.4	52.9	41.3	31.6		
	DBStd	18.76	12.58	11.64	11.41	9.73	8.71	6.67	5.34	5.26	7.26	9.09	9.87	10.55		
	HDD50	2862	753	604	420	126	14	0	0	0	1	73	295	576		
	HDD65	6056	1214	1018	856	471	205	39	7	12	98	388	712	1036		
	CDD50	2999	4	5	30	111	325	564	682	641	433	165	33	6		
	CDD65	717	0	0	1	6	52	153	224	188	79	14	0	0		
	CDH74	5557	0	0	9	59	432	1189	1793	1362	622	90	1	0		
	CDH80	1495	0	0	1	4	87	334	534	354	168	13	0	0		
(14) Wind	WSAvg	9.0	11.1	10.5	10.3	10.4	8.8	7.7	6.8	6.5	7.1	8.8	10.0	10.2		
(15) Precipitation	PrecAvg	39.8	2.3	2.0	3.2	3.7	4.3	4.0	4.1	4.0	3.3	2.4	3.5	2.9		
	PrecMax	67.2	5.6	4.7	7.0	6.6	8.8	10.0	13.2	7.6	7.8	5.7	12.8	11.2		
	PrecMin	21.8	0.4	0.3	1.2	0.7	1.5	0.6	0.9	0.6	0.7	0.4	0.7	0.7		
	PrecStd	9.0	1.1	1.1	1.5	1.6	1.8	2.2	2.4	1.8	1.8	1.2	2.3	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	64.7	73.1	80.1	85.6	90.0	91.3	89.8	88.9	81.5	69.7	61.4		
		MCWB	56.2	57.0	59.4	62.7	69.1	72.9	75.9	74.4	71.4	67.6	58.2	56.7		
	2%	DB	55.6	56.4	66.3	75.0	82.5	86.8	88.1	87.0	84.9	76.1	64.3	56.4		
		MCWB	53.0	51.6	56.2	60.0	67.3	71.5	73.8	73.0	70.1	63.9	56.6	52.6		
	5%	DB	50.0	51.0	61.2	70.9	79.2	84.0	85.5	84.0	81.0	72.1	60.8	51.8		
		MCWB	47.2	46.2	53.3	58.2	65.7	70.3	72.5	71.5	68.4	62.1	53.4	48.4		
	10%	DB	43.6	45.1	55.7	66.2	75.1	81.1	82.9	81.3	77.1	67.7	56.7	46.4		
		MCWB	40.3	41.1	49.1	55.4	64.1	68.7	70.8	69.9	65.9	60.0	50.8	42.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.9	58.3	61.7	66.4	72.6	75.9	78.4	77.6	75.2	71.0	62.4	58.2		
		MCDB	59.0	63.7	70.1	75.2	82.4	85.8	88.1	86.1	85.6	78.2	67.1	60.7		
	2%	WB	53.3	52.5	57.7	62.6	70.0	73.7	76.2	75.3	72.6	67.3	58.1	53.4		
		MCDB	55.5	55.4	64.4	71.5	78.8	83.0	85.0	83.3	80.4	72.9	62.7	55.2		
	5%	WB	47.6	46.4	54.5	59.9	67.9	72.2	74.5	73.6	70.6	64.0	54.9	48.6		
		MCDB	49.5	50.1	60.1	67.5	76.1	81.1	82.3	80.2	77.1	69.9	59.3	51.3		
	10%	WB	40.5	41.2	50.3	57.1	65.7	70.6	72.9	72.0	68.4	60.8	51.3	43.4		
		MCDB	43.6	45.0	55.8	64.7	73.4	78.5	80.0	78.3	74.6	66.7	56.1	46.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.8	14.5	16.6	20.1	19.5	18.7	18.7	18.4	19.4	18.0	14.8	12.5		
		MCDBR	17.4	20.7	23.3	25.0	23.1	21.4	20.5	20.3	22.5	22.6	19.6	17.6		
		MCWBR	15.8	17.2	15.2	14.5	11.6	9.7	9.0	9.0	10.2	12.2	13.8	15.4		
	5% WB	MCDBR	16.9	19.8	20.8	21.7	19.6	18.6	17.8	17.2	18.4	19.5	17.6	17.6		
		MCWBR	15.9	17.7	14.8	14.7	10.8	9.5	8.7	8.9	9.4	11.9	14.1	16.7		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.325	0.348	0.386	0.419	0.438	0.443	0.426	0.378	0.355	0.320	0.302		
	taud		2.420	2.332	2.354	2.270	2.213	2.199	2.203	2.235	2.351	2.403	2.467	2.445		
	Ebn at Noon		273	280	286	281	273	267	264	266	272	266	263	262		
	Edh at Noon		25	32	35	41	45	45	45	42	35	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		473	743	1077	1463	1729	1897	1890	1703	1374	916	595	400		
	RadStd		46	98	101	127	145	136	111	75	121	75	86	47		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.94	N/A	N/A	N/A	N/A	N/A	N/A	-225	+236	+131
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+205	+112

Nomenclature: See separate page